

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1: Experiments**

**COURSE NAME:** Cloud Computing and **COURSE CODE:** CSA15  
Big Data Analytics

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**COURSE OUTCOME COVERED**

**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for realworld applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)  
Question Count: 4

**Total Marks: 40**

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**Code of Conduct**

I Athavan k(192524036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Athavan k

### Experiments Rubric

| Criteria                 | Weightage | Level 4 - Exemplary                                   | Level 3 - Proficient              | Level 2 - Developing   | Level 1 - Beginning             |
|--------------------------|-----------|-------------------------------------------------------|-----------------------------------|------------------------|---------------------------------|
| Experimental Design      | 25%       | Design is systematic and technically strong           | Mostly correct design             | Basic design with gaps | Poorly designed activity        |
| Use of Tools / Platforms | 20%       | Appropriate and effective use of cloud tools          | Good tool usage with minor issues | Limited tool usage     | Incorrect or missing tool usage |
| Application of Concepts  | 20%       | Concepts are applied accurately to practical tasks    | Mostly accurate application       | Partial application    | Weak application                |
| Analysis and Output      | 20%       | Strong analysis and meaningful outcome interpretation | Good analysis with minor gaps     | Basic analysis         | Little or no analysis           |
| Documentation            | 15%       | Clear, structured, and complete documentation         | Mostly complete documentation     | Partially complete     | Poor documentation              |

Total Marks Awarded:

## Assessment Tasks

**Experiment 1:** Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

**Experiment 2:** Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

**Experiment 3:** Demonstrate virtualization by Installing Type2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

**Experiment 4:** Demonstrate virtualization by Installing Type2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

## **Experiment 1: Mark Sheet Processing System (Zoho Creator - SaaS)**

### **Aim**

To design and develop a simple cloud-based Software-as-a-Service (SaaS) application named “Mark Sheet Processing System” using Zoho Creator, a Platform-as-a-Service cloud tool, that records student and subject-wise mark details and automatically computes the average, percentage, rank and overall performance of each student.

### **Procedure**

1. Sign in to Zoho Creator ([creator.zoho.com](https://creator.zoho.com)) using a valid Zoho account.
2. Click “Create New Application” → “Create from scratch” and name the application “Mark Sheet Processing System”.
3. Inside the application, create a new Form and name it “Student Mark Sheet”.
4. Add the student detail fields: Student Name, Register Number, Department (Single Line) and Semester (Dropdown).
5. Add subject-wise fields for Subject 1 to Subject 5, each capturing the subject name and marks obtained.
6. Add formula fields (using Deluge script) for Total Marks (sum of all subject marks) and Average (Total Marks ÷ Number of subjects).
7. Add a formula field for Percentage and a rule-based field for Overall Performance (e.g., Excellent/Good/Average based on percentage range).
8. Add a Rank field that is updated by comparing the percentage of all submitted records.
9. Set mandatory-field validation and a numeric range (0–100) validation on the marks fields.
10. Save the form, click Preview/Deploy and fill sample student data to test the form end-to-end.
11. Submit the form and verify that Total Marks, Average, Percentage, Rank and Overall Performance are computed automatically.
12. Open the default Report linked to the form to view all submitted mark sheets in a grid (list) view.
13. Click on any record in the report to open its Detail View and confirm the values are stored and displayed correctly.
1. Publish the application so that it is accessible as a live SaaS link, completing the cloud deployment.

## Sample Screenshots

Mark Sheet Processing System

**Student Details**

Student Name \*

Register Number \*

Department \*

Semester \*

**Subject-wise Marks**

Subject 1 - Name & Marks \*

Subject 2 - Name & Marks \*

Subject 3 - Name & Marks \*

Subject 4 - Name & Marks \*

Subject 5 - Name & Marks \*

*Input – Zoho Creator form: Student Details and Subject-wise Marks fields*

Mark Sheet Processing System

**Result Summary (Auto Calculated)**

Total Marks

Average

Percentage

Rank

Overall Performance \*

*Input – Zoho Creator form: auto-calculated Result Summary and Submit*

| Mark Sheet Report                                                                                   |             |             |         |            |      |             |
|-----------------------------------------------------------------------------------------------------|-------------|-------------|---------|------------|------|-------------|
| <div> <input type="text"/> <input type="button" value="+"/> <input type="button" value="⋮"/> </div> |             |             |         |            |      |             |
| Student Name                                                                                        | Reg No      | Total Marks | Average | Percentage | Rank | Performance |
| Arun Kumar                                                                                          | 3122210xxxx | 430/500     | 86.0    | 86%        | 1    | Excellent   |
| Divya Sri                                                                                           | 3122210xxxy | 398/500     | 79.6    | 79.6%      | 2    | Good        |
| Karthik R                                                                                           | 3122210xxxz | 360/500     | 72.0    | 72%        | 3    | Good        |

*Output – Mark Sheet Report (grid view of submitted records)*

Student Name

Arun Kumar

Register Number

3122210xxxx

Department

CSE

Semester

V

Total Marks

430 / 500

Average

86.0

Percentage

86%

Rank

1

Overall Performance

Excellent

*Output – Detail view of a single student mark sheet record*

## **Experiment 2: Property Buying & Rental Process – Chennai City (Zoho Creator - SaaS)**

### **Aim**

To design and develop a simple cloud-based SaaS application named “Property Buying & Rental System” using Zoho Creator to record and manage property Buy / Rent / Commercial listings across Chennai city, including area, rental agreement status, budget range and property value.

### **Procedure**

2. Sign in to Zoho Creator and create a new application named “Property Buying & Rental System”.
3. Create a new Form named “Property Details” inside the application.
4. Add a Radio field “Transaction Type” with options Buy, Rent and Commercial.
5. Add a Dropdown field “Property Type” with options such as Apartment, Villa, Plot and Commercial Space.
6. Add a Dropdown field “Place” listing Chennai localities (e.g., Velachery, T Nagar, OMR, Adyar) and a Number field “Area (Sq.ft)”.
7. Add a Radio field “Rental Agreement” (Yes/No) and configure a Deluge “show/hide” rule so it is visible only when Transaction Type = Rent.
8. Add a Dropdown field “Range” for the budget bracket and a Currency/Number field “Property Value”.
9. Add “Owner Name” and “Contact Number” fields, applying a 10-digit regex validation on the phone number.
10. Save the form, deploy it, and test that the Rental Agreement field appears/disappears correctly based on Transaction Type.
11. Submit sample records for Buy, Rent and Commercial property categories to test all field combinations.
12. Open the associated Report to view the Property Listing Report in a grid (list) view.
13. Click a record to open its Detail View and verify that all captured values are accurate.
14. Publish the application to make it accessible as a live SaaS application, completing the cloud deployment.

### **Sample Screenshots**

Property Buying & Rental - Chennai

Property Category

Transaction Type \*

☐ Buy
 ☒ Rent
 ☐ Commercial

Property Type \*

Apartment ▾

Place \*

Velachery, Chennai ▾

Area (Sq.ft) \*

1250

*Input – Zoho Creator form: Property Category (Transaction Type, Property Type, Place, Area)*

Property Buying & Rental - Chennai

Agreement & Value Details

Rental Agreement \*

☒ Yes
 ☐ No

Range (Budget) \*

₹15,000 - ₹20,000 / month ▾

Property Value (₹) \*

18,500

Owner Name \*

Contact Number \*

Submit

*Input – Zoho Creator form: Agreement & Value details and Submit*

| Transaction Type | Property Type | Place     | Area (Sq.ft) | Rental Agreement | Value (₹) |
|------------------|---------------|-----------|--------------|------------------|-----------|
| Rent             | Apartment     | Velachery | 1250         | Yes              | 18,500/mo |
| Buy              | Villa         | OMR       | 2400         | -                | 1.8 Cr    |
| Commercial       | Office Space  | T Nagar   | 900          | Yes              | 45,000/mo |

*Output – Property Listing Report (grid view of submitted records)*

|                           |  |
|---------------------------|--|
| Duplicate                 |  |
| Transaction Type          |  |
| Rent                      |  |
| Property Type             |  |
| Apartment                 |  |
| Place                     |  |
| Velachery, Chennai        |  |
| Area (Sq.ft)              |  |
| 1250                      |  |
| Rental Agreement          |  |
| Yes                       |  |
| Range (Budget)            |  |
| ₹15,000 - ₹20,000 / month |  |
| Property Value            |  |
| ₹18,500 / month           |  |
| Owner Name                |  |
| S. Kalaivani              |  |

*Output – Detail view of a single property record*

### **Experiment 3: Virtualization – Create a VM (5 CPU, 12 GB RAM, 8 GB Storage) using a Type-2 Hypervisor**

#### **Aim**

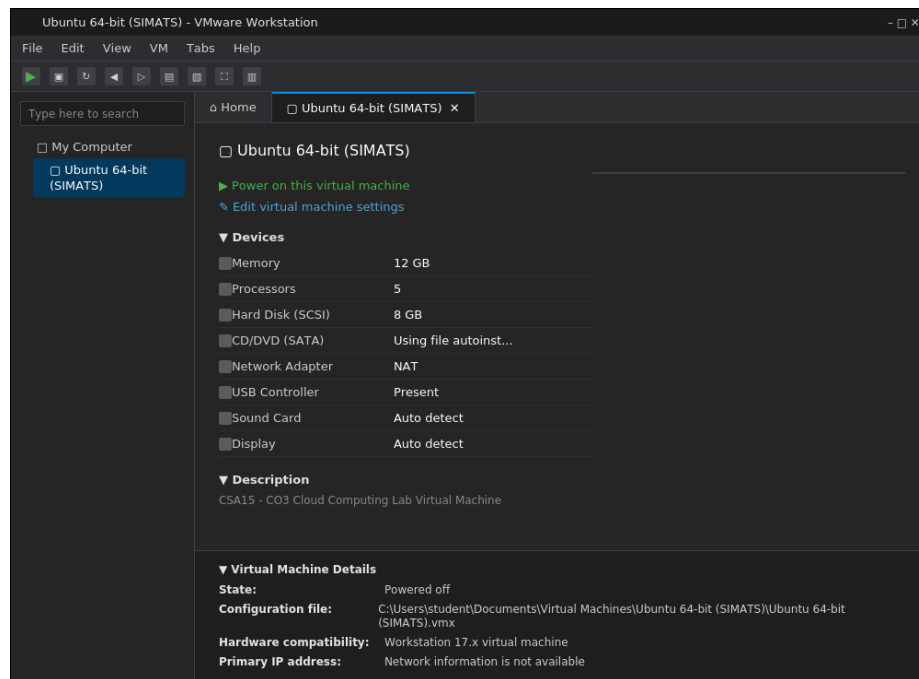
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation) on the host device, and to create and configure a Virtual Machine with a Host Operating System (Windows/Linux) allocated 5 CPU cores, 12 GB RAM and an 8 GB storage disk.

#### **Procedure**

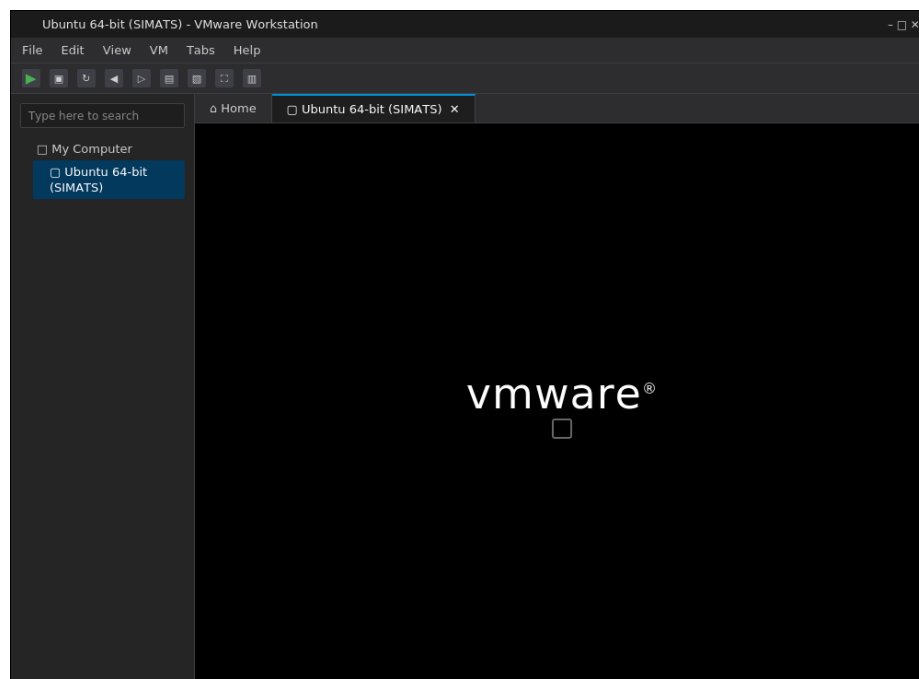
15. Download the VMware Workstation Pro/Player installer from the official VMware website, matching the host OS (Windows/Linux).
16. Install VMware Workstation on the host machine by running the installer and accepting the license agreement.
17. Launch VMware Workstation and click “Create a New Virtual Machine” to start the New Virtual Machine Wizard.
18. Select “Typical” or “Custom” configuration and choose the installer source (installation ISO or “I will install the operating system later”).
19. Select the Guest Operating System type and version (Windows/Linux) to be hosted on the VM.
20. Specify the Virtual Machine name (e.g., SIMATS-Lab-VM) and choose the storage location on the host disk.
21. In the Processor Configuration step, set the “Number of processors” to 5.
22. In the Memory configuration step, allocate Memory = 12 GB to the virtual machine.
23. In the Disk configuration step, set the Maximum disk size to 8 GB and choose “Store virtual disk as a single file”.
24. Review the configuration summary (5 CPU, 12 GB RAM, 8 GB disk) and click “Finish” to create the VM.
25. Power on the Virtual Machine and complete the guest operating system installation.
26. Verify the allocated hardware resources from within the guest OS (System Properties/Task Manager) to confirm 5 vCPUs, 12 GB RAM and 8 GB disk are correctly assigned.

#### **Sample Screenshots**

## *Step 1 – Memory and Disk Configuration: 12 GB RAM and 8 GB disk*



## *Step 2 – VMware Workstation Library: VM summary showing Memory 12 GB, Processors 5, Hard Disk 8 GB*



## *Step 3 – Virtual machine powered on (VMware boot screen) with the Ubuntu guest OS starting*

The screenshot shows a VMware Workstation window titled "Ubuntu 64-bit (SIMATS) - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help) and a toolbar with various VM controls. On the left, a sidebar shows a search bar and a list of virtual machines under "My Computer", with "Ubuntu 64-bit (SIMATS)" selected. The main area is a terminal window with the following output:

```
student@ubuntu:~$ nproc
5
student@ubuntu:~$ free -h
              total        used        free
Mem:          11Gi       1.2Gi       9.8Gi
student@ubuntu:~$ lsblk
sda      8:0  0  8G  0 disk
student@ubuntu:~$ 5 vCPU / 12GB RAM / 8GB Disk confirmed
```

*Step 4 – Verifying allocated resources inside the guest OS using nproc, free -h and lsblk*

## **Experiment 4: Virtualization – Snapshot and Clone of a VM (Test with Previous/Cloned VM)**

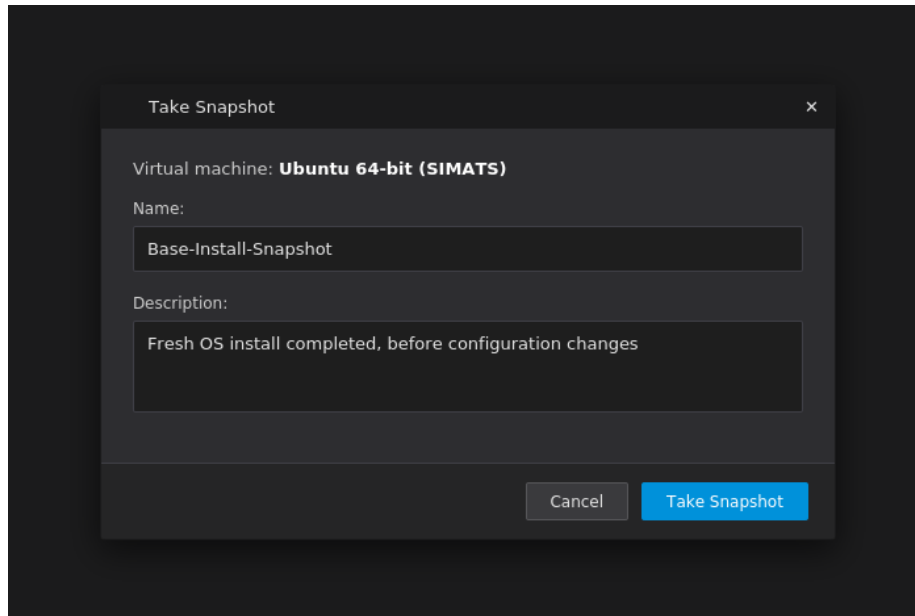
### **Aim**

To demonstrate virtualization by installing a Type-2 Hypervisor, creating and configuring a Virtual Machine with a Host Operating System, and then creating a Snapshot of the VM and a Clone of the VM, testing it by loading the previous version / cloned VM.

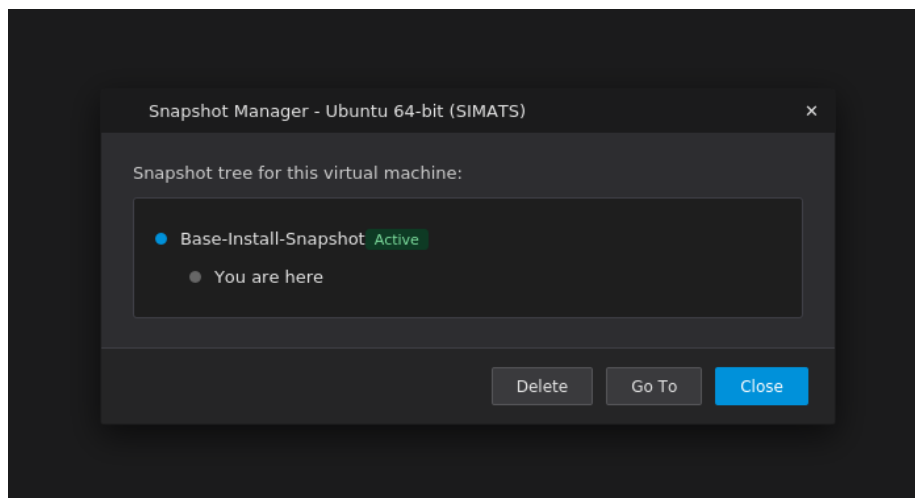
### **Procedure**

27. Ensure VMware Workstation (Type-2 Hypervisor) is installed and the base virtual machine created earlier is available and powered on.
28. Go to the VM menu → Snapshot → Take Snapshot to open the Take Snapshot dialog.
29. Enter a Snapshot Name (e.g., Base-Install-Snapshot) and an optional Description, then click “Take Snapshot” to capture the current VM state.
30. Open VM → Snapshot → Snapshot Manager to view the snapshot tree and confirm that the snapshot has been created successfully.
31. Make a change on the VM (e.g., install an application or modify a setting) so the current state differs from the snapshot.
32. Go to VM menu → Manage → Clone to open the Clone Virtual Machine Wizard.
33. Under Clone Source, select “An existing snapshot” and choose the snapshot created in Step 3.
34. Under Clone Type, choose “Full clone” (independent copy) or “Linked clone” (references the snapshot) as required.
35. Provide a name and destination location for the cloned VM (e.g., SIMATS-Lab-VM-Clone) and click “Finish”.
36. Power on the cloned VM and verify that it boots correctly, reflecting the state captured in the snapshot.
37. To test the previous version on the original VM, open Snapshot Manager, select the snapshot and click “Go To/Restore” to revert the VM to that saved state.
38. Confirm that the reverted VM matches the previous (snapshot) version, validating that both the snapshot and clone operations were successful.

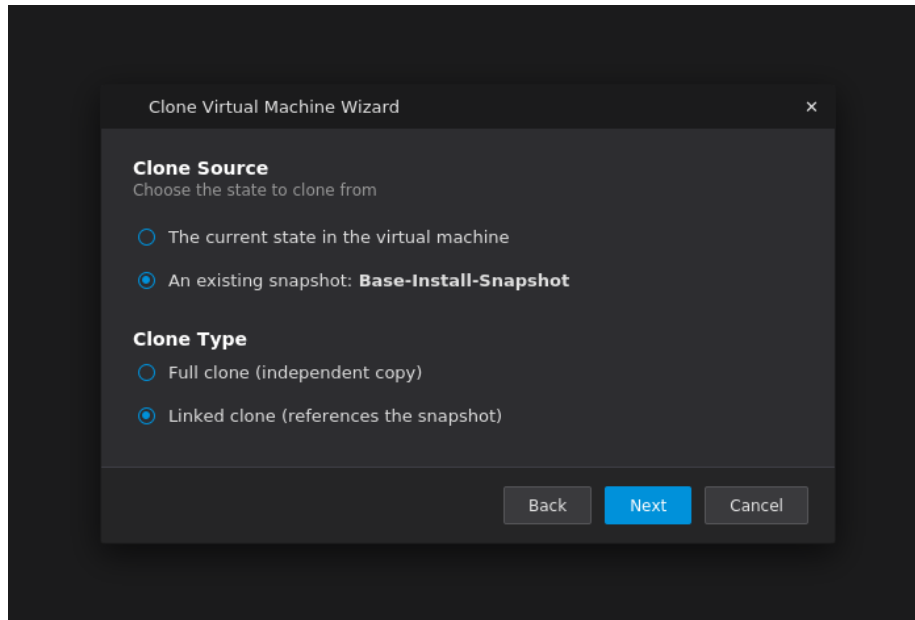
### **Sample Screenshots**



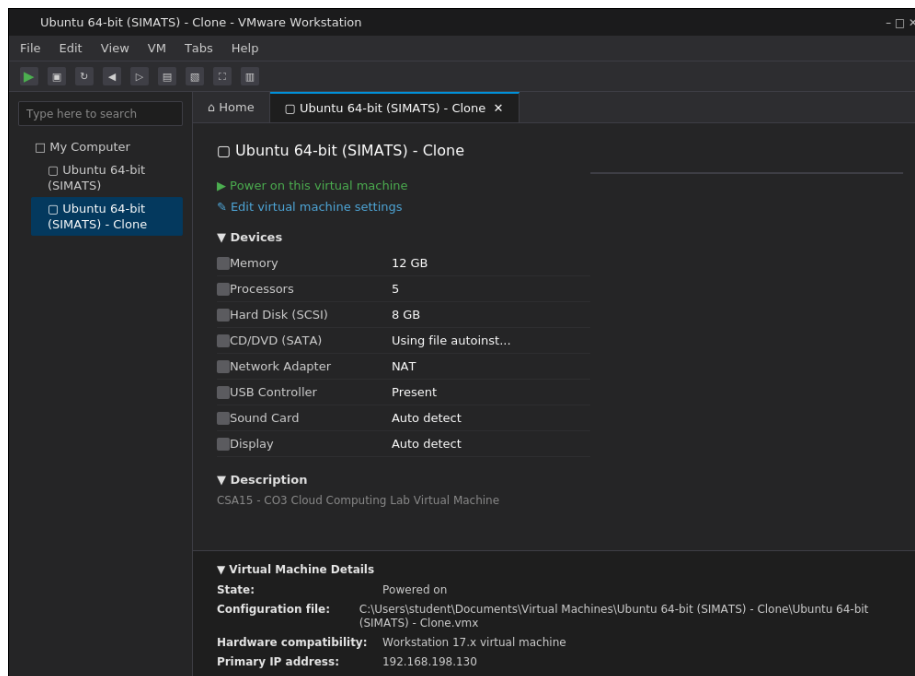
*Step 1 – Take Snapshot dialog with name and description*



*Step 2 – Snapshot Manager showing the created snapshot*



*Step 3 – Clone Virtual Machine Wizard: source snapshot and clone type*



*Step 4 – Cloned VM in the Library, powered on and verified*